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3-dimensionally printed patient-specific glenoid drill guides vs. standard nonspecific instrumentation: a randomized controlled trial comparing the accuracy of glenoid component placement in anatomic total shoulder arthroplasty

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Background: Traditional, commercially sourced patient-specific instrumentation (PSI) systems for shoulder arthroplasty improve glenoid component placement but can involve considerable cost and outsourcing delays. The purpose of this randomized controlled trial was to compare the accuracy of glenoid component positioning in anatomic total shoulder arthroplasty (aTSA) using an in-house, point-of-care, 3-dimensionally (3D) printed patient-specific glenoid drill guide vs. standard nonspecific instrumentation.
Methods: This single-center randomized controlled trial included 36 adult patients undergoing primary aTSA. Patients were blinded and randomized 1:1 to either the PSI or the standard aTSA guide groups. The primary endpoint was the accuracy of glenoid component placement (version and inclination), which was determined using a metal-suppression computed tomography scan taken between 6 weeks and 1 year postoperatively. Deviation from the preoperative 3D templating plan was calculated for each patient. Blinded postoperative computed tomography measurements were performed by a fellowship-trained shoulder surgeon and a musculoskeletal radiologist.
Results: Nineteen patients were randomized to the patient-specific glenoid drill guide group, and 17 patients were allocated to the standard instrumentation control group. There were no significant differences between the 2 groups for native version ($P = .527$) or inclination ($P = .415$). The version correction was similar between the 2 groups ($P = .551$), and the PSI group was significantly more accurate when correcting version than the control group ($P = .042$). The PSI group required a significantly greater inclination correction than the control group ($P = .002$); however, the 2 groups still had similar accuracy when correcting inclination ($P = .851$). For the PSI group, there was no correlation between the accuracy of component placement and native version, native inclination, or the Walch

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The Rush University Medical Center institutional review board approved this study (approval #18020102).

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classification of glenoid wear ($P > .05$). For the control group, accuracy when correcting version was inversely correlated with native version ($P = .033$), but accuracy was not correlated with native inclination or the Walch classification of glenoid wear ($P > .05$). The intraclass correlation coefficient was 0.703 and 0.848 when measuring version and inclination accuracy, respectively.
Conclusion: When compared with standard instrumentation, the use of in-house, 3D printed, patient-specific glenoid drill guides during aTSA led to more accurate glenoid component version correction and similarly accurate inclination correction. Additional research should examine the influence of proper component position and use of PSI on clinical outcomes.
Level of evidence: Level I; Randomized Controlled Trial; Treatment Study

Keywords: Patient-specific instrumentation; anatomic total shoulder arthroplasty; glenoid drill guides; 3D printing; 3D templating; preoperative planning

Accurate glenoid component implantation during anatomic total shoulder arthroplasty (aTSA) is one of the most challenging aspects of the procedure due to unique, patient-specific variations in glenoid version, inclination, and bone loss.[3,4,14] Surgical precision and accuracy during TSA is clinically significant, as poorly placed glenoid components can cause increased stress at the bone-implant interface and lead to an increased rate of osteolysis.[2,7,9,13] Accurate component placement that is specific to a patient's anatomy leads to reduced rates of wear, implant loosening, osteolysis, and failure over time.[2,12] Despite substantial improvements to implant design and surgical technique, the intraoperative implementation of a preoperative plan using standard instrumentation remains intrinsically imprecise as the surgeon must correct for multiplanar deformities while only visualizing a small part of the scapula—the glenoid face.[8,11,13,18,25]

Patient-specific instrumentation (PSI) has emerged as a promising solution to improve accuracy, precision, and efficiency during TSA.[2] Patient-specific glenoid drill guides are designed to assist in placement of the central glenoid guide pin, which is used to control glenoid reaming and direct accurate placement of the glenoid component.[4,6,10,24] This allows for accurate and reliable intraoperative implementation, particularly when combined with advanced software for preoperative planning.[21] Thus far, there are substantial preclinical data demonstrating the improved accuracy PSI provides during glenoid component placement for TSA.[6,10,16,21,24] Though promising, there still remains a relative paucity of comparative clinical studies demonstrating the efficacy of PSI in TSA with only 1 published randomized controlled trial (RCT) to date.[8,23]

Although traditional, commercially sourced PSI systems for TSA significantly improve the accuracy of glenoid component placement, they involve considerable cost and outsourcing time delays that make them challenging for widespread adoption.[2,22] For PSI to be regularly used in TSA, the development of efficient and economic glenoid drill guides would be valuable.[2] A prior cadaveric study by our group demonstrated that a novel in-house, 3-dimensionally (3D) printed patient-specific glenoid drill guide had improved accuracy relative to traditional, nonspecific TSA instrumentation (Fig. 1).[2] Although ergonomic, these point-of-care, 3D printed PSI guides currently lack clinical data. In addition, there is no prior RCT that directly compares PSI guides with controls treated with 3D computed tomography (CT) preoperative templating.

The purpose of this trial was to compare the accuracy of glenoid component positioning between custom 3D printed patient-specific glenoid drill guides and standard nonspecific instrumentation with 3D CT preoperative templating during aTSA. We hypothesized that the PSI group would demonstrate superior accuracy relative to the control group in regard to correcting glenoid version.

Materials and methods

Trial design

This single-center, blinded, prospective RCT was conducted at Midwest Orthopaedics at Rush and RUSH University Medical Center between June 2018 and May 2022. The hypothesis being tested was developed before patient enrollment, and the Consolidated Standard of Reporting Trials (CONSORT) guidelines were followed to perform this study (Fig. 2). This RCT received no external funding. All subjects provided written informed consent before enrollment, blinding, and randomization. All patients undergoing an aTSA during the study timeline were identified and assessed for eligibility. Patients were included if they were between the ages of 18 and 80 years, underwent a primary aTSA, had a preoperative CT scan of the affected shoulder, provided written consent, and were willing to undergo a postoperative CT scan of their shoulder within 6 weeks to 1 year of the index procedure. Patients were excluded if they had a previous aTSA or hemiarthroplasty, are currently pregnant, or had a past medical history of cancer, coagulopathy, inflammatory musculoskeletal condition, or connective tissue disorder. Included patients were consented by 2 nonsurgeon investigators (Z.A.K. and A.S.V.) and randomized using a random number generator to minimize randomization bias. Patients were blinded to their assigned treatment arm.

Sample size

There were 2 cohorts of patients who underwent aTSA by fellowship-trained, board-certified orthopedic surgeons (NNV, GEG, GPN, BJC, and BF). Previously published arthroplasty

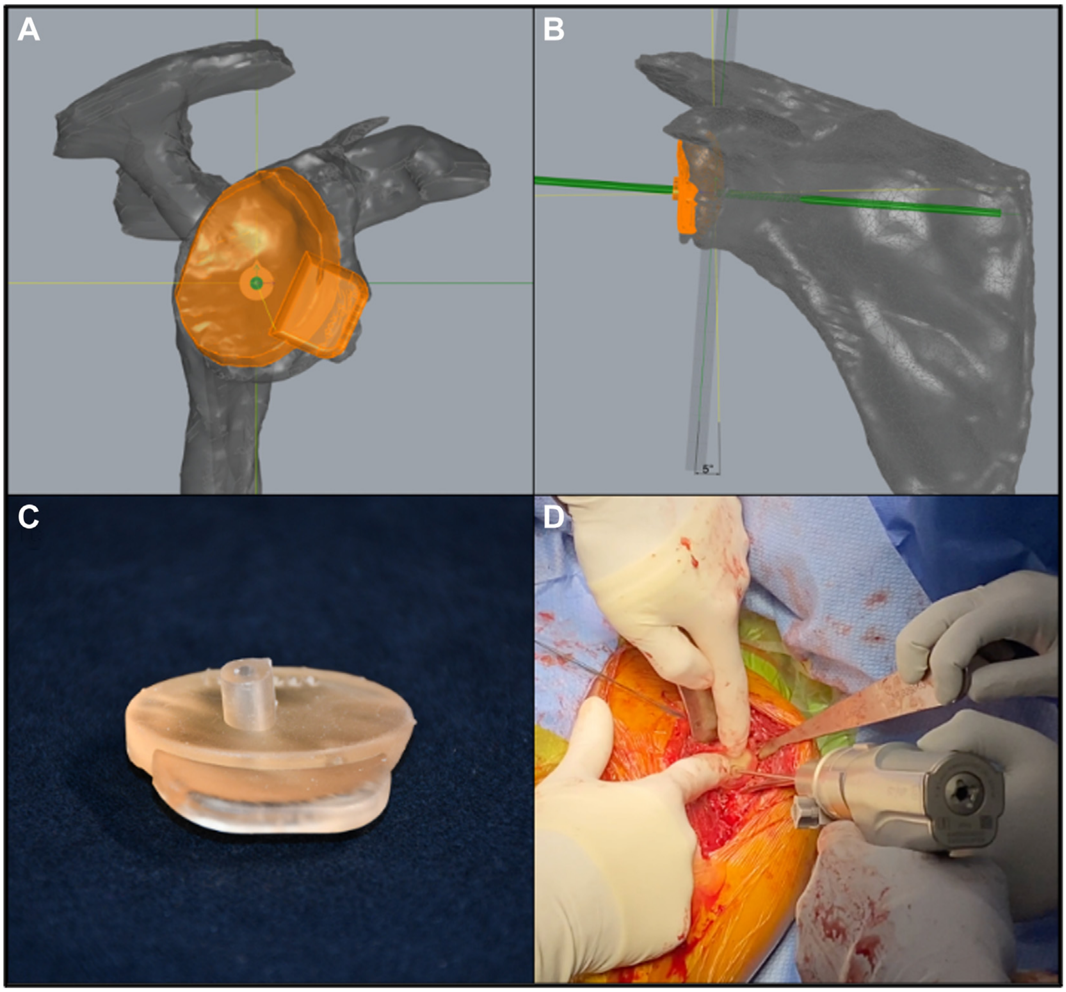

Figure 1 Preoperative design of a patient-specific instrumentation guide from a view in the sagittal plane that is perpendicular to the centroid-trigonum axis (**A**) and a view in the coronal plane (**B**). (**C**) The barrel with an internal diameter of 2.75 mm. (**D**) Intraoperative placement of the guide during an anatomic total shoulder arthroplasty.

studies have demonstrated that a total of 26 patients (13 patients in each group) are needed to detect 1 degree of alignment difference at an α of 0.05 and a power of 0.8.[5,19,15] A prior RCT demonstrated a significant benefit to PSI in TSA with 31 enrolled patients.[8] Thus, the present RCT targeted an enrollment of total 40 patients (20 patients in each group) to sufficiently power the study and account for patients who could be lost to follow-up.

Patient enrollment

Forty patients were enrolled in the study, blinded, and randomized to either the PSI group or the standard instrumentation control group. Two patients did not schedule postoperative CT scans, so they were excluded from the analysis. Two patients did not proceed with an aTSA after enrollment and randomization. The remaining 36 patients were included in the analysis. All patients underwent preoperative CT scans on a Brightspeed scanner (General Electric, Waukesha, WI, USA) according to the following parameters: 0.625 mm slice thickness, 120 kV, 260 milliampere-seconds, and an image matrix of 512 $\times$ 512.[2]

Preoperative planning and surgical technique

The PSI group used a patient-specific glenoid drill guide that was described in detail by Cabarcas et al.[2] Briefly, preoperative 2D CT scan DICOMS (Digital Imaging and Communications in Medicine) were used to create 3D models using commercially available and Food and Drug Administration–approved segmentation software (Mimics Medical, version 24.0; Materialise, Leuven, Belgium). These models were later exported in the Standard Tessellation Language (.STL) file format.

Coordinate system definitions

The coronal scapular plane was defined by the center of the glenoid face, the trigonum, and the inferior angle of the scapula.[1] The axial scapular plane was established by creating a plane that is orthogonal to the coronal scapular plane.[19] The .STL files were imported into a custom-written Visual C++ program (Microsoft Foundation Class; Microsoft, Redmond, WA, USA), which implemented an algorithm to calculate the exact coordinates of the center of the glenoid face, trigonum, and inferior angle of the scapula.

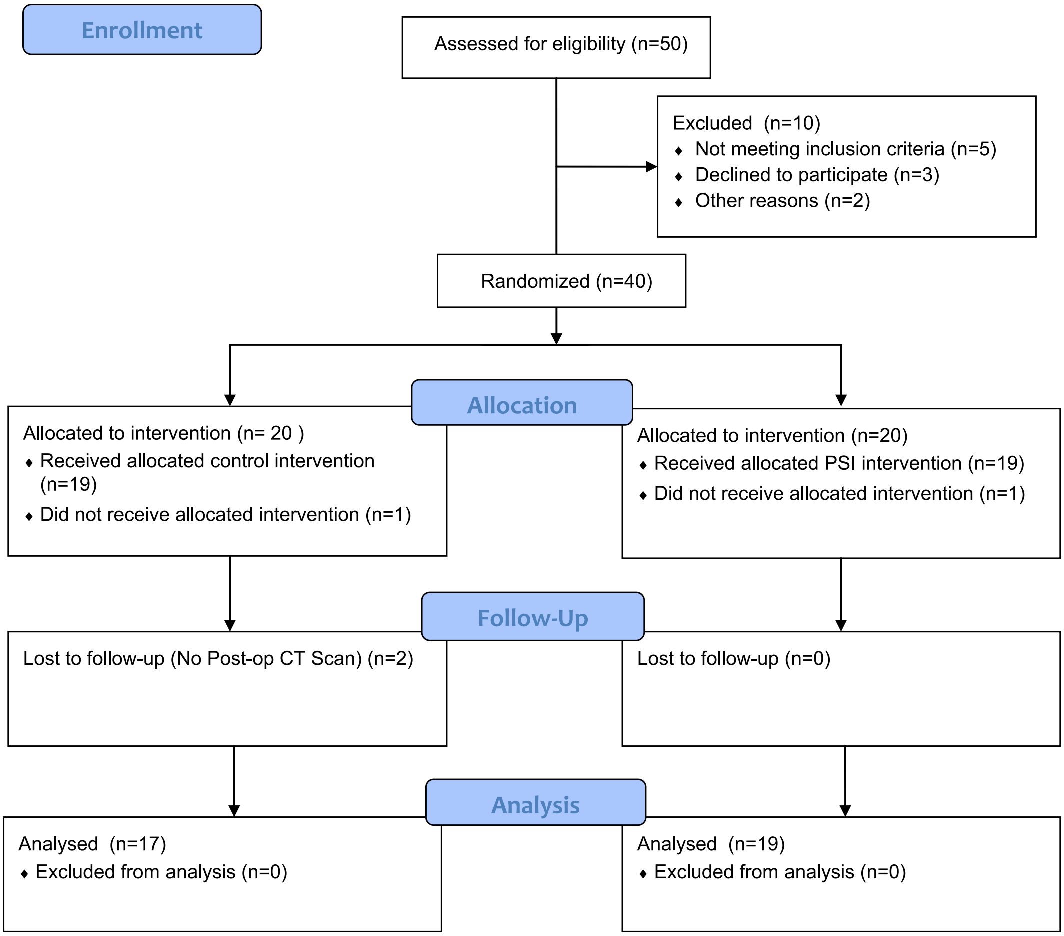

Figure 2 Consolidated Standard of Reporting Trials (CONSORT) flow diagram demonstrating inclusion and final group allocation for the control group (*left*) and the patient-specific instrumentation (PSI) group (*right*). *PSI*, patient-specific instrumentation; *Post-op CT*, postoperative computed tomography.

Guide design and manufacturing

After defining these 3 key coordinates, the 3D scapular model .STL file was imported into 3D computer-aided design software (Rhinoceros, version 7; Robert McNeel & Associates, Seattle, WA, USA) to design the PSI guide as a perfect reflection of the glenoid face topography. This allowed us to construct a novel PSI guide that fit snugly on the glenoid face, when oriented correctly (Fig. 3). The target version and inclination correction were obtained based on the treating surgeon's plan using 3D preoperative planning software that was specific to the patient and the implant used (Blueprint for Tornier/Stryker implants, Virtual Implant Positioning for Arthrex implants, and MatchPoint for DJO/enovis implants). A barrel with a 2.75 mm internal diameter was included on the PSI guide to drive a pin through the center of the glenoid face at the target angle provided by the preoperative template. The guide was printed with a stereolithography Formlabs Form 2 desktop 3D printer (Formlabs, Somerville, MA, USA) using Formlabs Biomed Amber Resin (class 1 biocompatible resin; EN-ISO 10993-1:2009/AC:2010, United States Pharmacopeia class VI). This resin is intended for the manufacture of surgical and pilot drill guides with short-term tissue contact. The guides were sterilized via autoclave (as recommended by the manufacturer) before each procedure and handled with standard sterile precautions. During the procedure, after exposing the glenoid, the patient-specific glenoid drill guide was placed on the glenoid face. It is designed to have an exact fit on the glenoid face. Tactile feedback from the perfect fit between the PSI guide and the glenoid along with a cranially pointing arrow confirms the proper orientation of the guide. This design feature establishes the correct location and drilling angle for glenoid pin placement.

The control group underwent a procedure using nonspecific standard instrumentation that is currently the standard of care for aTSA. As with the PSI group, all devices were implanted using a cannulated technique for the preparation of the glenoid face. Also equivalent to the PSI group, the same preoperative planning process was undertaken by the surgeon using the implant-specific software. The target version and inclination correction were again obtained using the software template system. A freehand technique using the standard guides available in the implant system was used to place the glenoid pin (Fig. 3). The estimated center of the glenoid vault was marked using electrocautery, and the glenoid guide pin was driven through the face of the glenoid using a standard guide. The surgeon's finger was placed over the anterior glenoid vault to "Matsen's point" to guide version and inclination

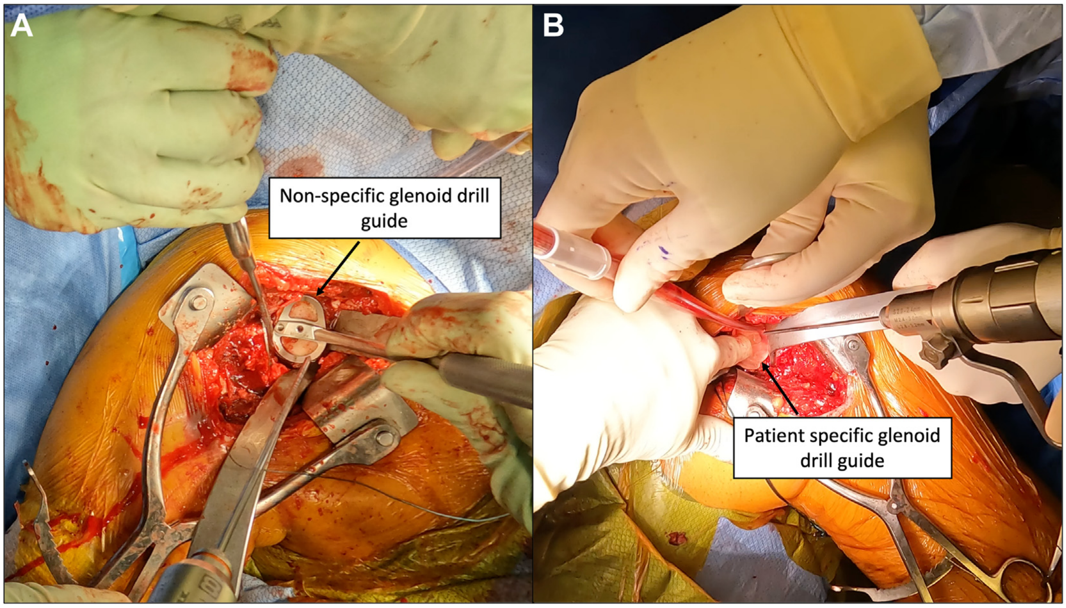

Figure 3 Example of translating our 3-dimensional (3D) preoperative templating plan using a standard nonspecific glenoid guide provided by the implant manufacturer (**A**). Example of using a 3D printed patient-specific glenoid drill guide that drives the pin at our target component version and inclination (**B**). The patient-specific guide is designed to have an exact fit on the glenoid face. Tactile feedback from the perfect fit between the patient-specific guide and the glenoid, as well as a cranially pointing arrow, confirms the proper orientation of the guide before drilling.

of the pin trajectory.[17] Beyond the design and intraoperative implementation of a 3D printed patient-specific glenoid drill guide, there were no differences in the preoperative planning or manufacturer-specific 3D templating between the control group and the PSI treatment group.

For both groups, all surgical procedures were performed by surgeons with extensive experience in TSA (NNV, GEG, GPN, BJC, and BF). The available implants were the same for all surgeons across both groups. The surgeons participating in this RCT each used 1 of the 3 companies listed above for aTSA implants and 3D preoperative planning (Tornier Blueprint, Arthrex VIP, or DJO MatchPoint). As a result, details related to the exact implant-specific blueprints for preoperative planning or manufacturer-specific steps for surgical implantation of the glenoid component were beyond the scope of this paper. Briefly, a deltopectoral approach was used followed by a subscapularis peel, tenotomy, or lesser tuberosity osteotomy according to surgeon preference. After preparing the humerus, attention was turned to the glenoid, where it was exposed and prepared. The glenoid pin was placed using either a patient-specific glenoid drill guide for the PSI group or using a freehand technique with a nonspecific guide for the control group. The guide pin was used to control the reamer and direct glenoid component placement. Other than the drill guide used to direct placement of the guide pin, there were no differences between the 2 groups during the surgical procedure or their subsequent postoperative care.

Primary clinical outcomes and data analysis

The primary endpoint was the accuracy of glenoid component placement relative to the planned target inclination and version provided by the surgeon's preoperative plan. Deviation from the preoperative plan was calculated for each patient. This was determined using a postoperative metallic artifact suppression CT scan taken between 6 weeks and 1 year after the index procedure (Fig. 4). A shoulder fellowship-trained orthopedic surgeon (MEM) and a board-certified musculoskeletal radiologist (JWE) were blinded to the patient treatment group and measured the final glenoid component version and inclination using this postoperative CT scan. The large range of timing of the postoperative CT scan was secondary to limitations with patient access during the COVID-19 pandemic.

Baseline characteristics were compared between the 2 cohorts. This included age, sex, laterality, subscapularis management, the Walch classification of glenoid wear, native inclination, native version, the use of a wedge augment, the targeted degrees of version correction, and the targeted degrees of inclination correction. The primary outcome for this study was the accuracy of glenoid component placement between the 2 groups. In this study, the accuracy of glenoid component placement was quantified using absolute error. Absolute error was determined by comparing the version/inclination of the final component placement as measured by a shoulder fellowship-trained orthopedic surgeon and board-certified musculoskeletal radiologist with the preoperative target version/inclination as determined by the implant-specific 3D templating software. The Walch classification was reported by the implant-specific templating blueprint and confirmed by a shoulder fellowship-trained orthopedic surgeon and board-certified musculoskeletal radiologist.

Statistical analysis was performed using SPSS Statistics (version 28.0; IBM Corp., Armonk, NY, USA). Continuous variables were presented as a mean $\pm$ standard deviation. Normality was assessed with a Shapiro-Wilk test before performing either an unpaired, 2-tailed t test or a Mann-Whitney U test. Categorical variables were represented with a count and compared using a

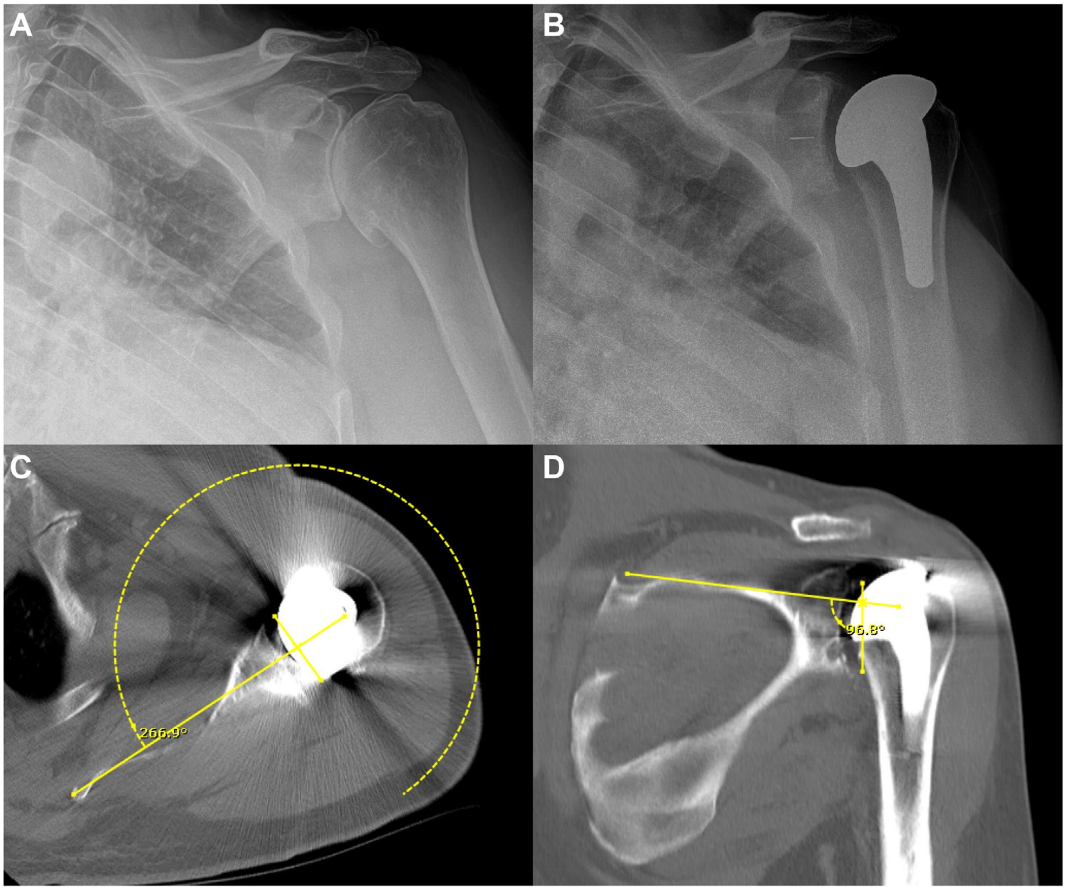

Figure 4 Preoperative radiograph demonstrating a patient with Walch classification B1 glenoid osteoarthritis (**A**) and corresponding postoperative radiograph (**B**). (**C** and **D**) Measurement of the patient's component version and inclination in their postoperative computed tomography scan. This patient had a native retroversion of 12° and a native superior inclination of 8°. The target correction for this patient was neutral version and 4° of superior inclination. This required the use of a wedge augment to achieve 12° of version correction and 4° of inclination correction. The patient's final component positioning was a version of −3.1° and an inclination of +6.8°. As a result, the patient had 3.1° of absolute error in the version correction and 2.8° of absolute error in the inclination correction.

Fisher's exact test. A Pearson's correlation coefficient was used to determine the relationship between inclination/version accuracy and native inclination/version. A Spearman's rank correlation coefficient was used to determine the relationship between the accuracy of component placement and the Walch classification of glenoid wear. An intraclass correlation coefficient (ICC) was calculated to determine inter-rater reliability for error in the correction of version and inclination. A 2-way mixed-model examining absolute agreement was implemented. An ICC greater than 0.7 indicated acceptable agreement between the 2 authors. Alpha was set at 0.05.

Results

Study population

Overall, there were no significant baseline differences between the 2 groups (Table I). Nineteen patients were randomized to the patient-specific glenoid drill guide group, and 17 were allocated to the standard instrumentation control group. Between the 2 groups, there were no differences in baseline age (P = .760), sex (P = 1.000), laterality (P = 1.000), subscapularis management (P = .819), or the Walch classification of glenoid wear (P = .974) (Tables I and II). The native glenoid version of the control group was −14.7° ± 8.4° (Table II). The native glenoid version of the PSI group was −12.9° ± 8.3°. This difference was not significant (P = .527) (Table II). The native glenoid inclination of the control group was 4.2° ± 3.6°, whereas the native glenoid inclination of the PSI group was 8.4° ± 4.2°. This difference was not significant (P = .415) (Table II).

Accuracy of component placement

Version correction for the PSI group was 12.2° ± 7.8° (range: 1.0°-27.8°), and version correction for the control group was 10.6° ± 7.6° (range: 0°-24.0°) (Table III). This difference was not significant (P = .551). The error in version correction for the PSI group was 3.1° ± 2.0°,

Table I Patient demographics

Treatment group	Control (n = 17)	Patient-specific instrumentation (n = 19)	*P* value
Age (yr), mean ± SD	61.0 ± 8.8	60.2 ± 7.7	.760
Sex			1.000
Female	5	6	
Male	12	13	
Laterality			1.000
Left	6	7	
Right	11	12	
Subscapularis management			.819
Peel	9	10	
LTO	6	5	
Tenotomy	2	4	

SD, standard deviation; *LTO*, lesser tuberosity osteotomy.

Table II Native glenoid morphology

Treatment group	Control (n = 17)	Patient-specific instrumentation (n = 19)	*P* value
Native inclination (°)	4.2 ± 6.1	8.4 ± 10.4	.415
Native version (°)	−14.7 ± 8.4	−12.9 ± 8.3	.527
Walch classification (n)			.974
A1	3	3	
A2	2	2	
B1	2	3	
B2	9	11	
B3	1	0	

whereas the error in version correction for the control group was 5.0° ± 3.3° (Table III). This difference was significant and indicated higher accuracy in the PSI group when correcting version during aTSA (P = .042).

Inclination correction for the PSI group was 6.9° ± 10.1° (range: 0°-36.5°), whereas inclination correction for the control group was 1.5° ± 1.5° (range: 0°-5°) (Table III). This difference was significant (P = .002). The error for correction of inclination in the PSI group was 9.5° ± 5.4°, and the error for correction of inclination in the control group was 9.6° ± 6.7° (Table III). Despite requiring a significantly greater correction, the PSI group had a similar accuracy to the control group when correcting inclination during aTSA (P = .851).

Correlation between accuracy and native glenoid wear

For the control group, the error in inclination was not related to native inclination (P = .547); however, the error in component placement with regard to version was significantly correlated with native version (P = .033) (Table IV). For the PSI group, there was no significant correlation between native version and accuracy (P = .114) or native inclination and accuracy (P = .493) (Table IV). For the control group, there was no correlation between the Walch classification of glenoid wear and accuracy for version (P = .354) or inclination (P = .384) (Table IV). For the PSI group, there was no correlation between the Walch classification of glenoid wear and accuracy for version (P = .312) or inclination (P = .114) (Table IV).

Inter-rater reliability

Inter-rater reliability between the 2 authors was determined using an ICC (Table V). The ICC for glenoid version accuracy measurements between the 2 authors was 0.703 (95% confidence interval: 0.214-0.820), indicating acceptable agreement between the 2 authors. The ICC for glenoid inclination accuracy measurements between the 2 authors was 0.848 (95% confidence interval: 0.692-0.975), indicating acceptable agreement between the 2 authors.

Discussion

This RCT demonstrated that (1) patient-specific glenoid drill guides were more accurate than the control technique with 3D preoperative templating when correcting version during aTSA, and although this finding was statistically significant, it is unclear if these results are clinically significant; (2) despite requiring a significantly greater

Table III Accuracy in glenoid component positioning

Treatment group	Control (n = 17)	Patient-specific instrumentation (n = 19)	*P* value
Correction for inclination (°)	**1.5 ± 1.5**	**6.9 ± 10.1**	**.002**
Correction for version (°)	10.6 ± 7.6	12.2 ± 7.8	.551
Absolute error in correction of inclination (°)	9.6 ± 6.7	9.5 ± 5.4	.851
Absolute error in correction of version (°)	**5.0 ± 3.3**	**3.1 ± 2.0**	**.042**

Bold values are statistically significant.

Table IV Correlations between accuracy and native glenoid bony erosion

Treatment group	Control (n = 17)	*P* value	Patient-specific instrumentation (n = 19)	*P* value
Correlation between native inclination and accuracy	**+0.518**	**.033**	+0.374	.114
Correlation between native version and accuracy	+0.157	.547	−0.167	.493
Correlation between the Walch classification of glenoid wear accuracy (inclination)	+0.240	.354	+0.245	.312
Correlation between the Walch classification of glenoid wear accuracy (version)	+0.128	.384	+0.375	.114

Table V Intraclass correlation coefficient (ICC)

Measurement	ICC [95% CI]
ICC for version accuracy	0.703 [0.214-0.820]
ICC for inclination accuracy	0.848 [0.692-0.975]

CI, confidence interval.

inclination correction, the patient-specific glenoid drill guides had equivalent accuracy to the control technique when correcting inclination during aTSA; (3) increased native retroversion was correlated with increased version error in the control group but was not correlated with increased version error in the PSI group; (4) there was no correlation between native glenoid inclination or the Walch classification of glenoid wear and accuracy of component placement for either group; and (5) there was acceptable inter-rater reliability between the 2 authors performing measurements, indicating validity of these results.

To date, there has only been 1 RCT that directly compared patient-specific glenoid drill guides with a conventional technique using standard instrumentation during aTSA.[8] In that study, Hendel et al[8] demonstrated that PSI was significantly more accurate when correcting inclination, but not version. However, the control group was treated using 2D CT imaging without 3D preoperative templating, and the authors targeted neutral correction for all control patients regardless of the patients' native glenoid morphology. Novel to our RCT study design, both groups received 3D preoperative templating with target inclination and version specific to the patient. Despite these differences, our investigation also demonstrated a statistically significant benefit for PSI relative to standard instrumentation when correcting version. Thus, in the context of the prior clinical literature, the results of this study would suggest that PSI improves the ability of a surgeon to intraoperatively implement 3D preoperative templating, leading to greater accuracy of glenoid component placement. However, it is critical to note that these results denote a statistically significant benefit, and it is unclear if this statistically significant benefit has a clinically significant effect on patient outcomes.

Prior clinical and cadaveric investigations have suggested that the benefit of PSI during aTSA is most evident in cases where there is more severe bony deformity.[8,12,21] These studies also demonstrated that the commercial PSI systems implemented became less accurate with increased glenoid bony erosion.[8,21] Surprisingly, for this investigation, there was no correlation observed between the accuracy of component placement and native inclination or Walch classification. It is possible that the use of 3D preoperative templating in the control group led to component placement that was not significantly impaired by increased inclination. However, unlike native inclination, increased native retroversion led to decreased accuracy in the control group. It can be difficult to visually judge retroversion in the setting of arthrosis and potentially limited exposure. Furthermore, anterior rim osteophytes may also conflate the degree of retroversion corrected with nonspecific instrumentation. Interestingly, the accuracy of the PSI group was not affected by either native version or native inclination. These findings would suggest that the in-house, 3D printed PSI glenoid drill guides do not become less effective when treating patients with severe bony deformity and further increases their clinical utility, particularly in cases with severe bony erosion and increased retroversion. A further study by our group is underway to determine the accuracy of the guide systems particularly in patients with severe glenoid deformity.

Although prior clinical investigations have consistently demonstrated superior accuracy in glenoid component placement during TSA when implementing PSI, they have largely used commercial PSI systems.[8,12,13] Commercially manufactured PSI systems in TSA have typically been reliable; however, they are associated with significant cost and production delays due to outsourcing that can make it challenging for the routine implementation of PSI in a clinical practice.[2] A prior study by our group reported on an ergonomic patient-specific guide that was preoperatively manufactured and designed to have a flush fit against the glenoid surface.[2] The average production time for the PSI guides from this study was approximately 5 hours per guide (including the design and printing time).[2] It takes approximately 1 hour to design and process a single guide and 4 hours to print a single guide, which can be done remotely. In addition, at our institution, up to 5 guides can be printed simultaneously. These guides are composed of a class 1 biocompatible resin that is designed to print surgical and pilot drill guides. This material is sterilized via autoclave at 134°C for 30 minutes and is safe for patient use as it remains in contact with the glenoid surface for less than 60 seconds during the procedure. In addition, these 3D printed custom guides demonstrated superior accuracy in a cadaveric model, making them safe, clinically effective, and time effective.[2] Furthermore, these guides are designed so that they require less extensive dissection and preparation of the glenohumeral joint relative to several other PSI guide models, which often require exposure of the coracoid base or acromion for adequate positioning.[2,12] The present investigation demonstrated that the statistically significant superior accuracy of these point-of-care manufactured PSI glenoid drill guides successfully translated to the clinical setting and remained effective in cases with severe glenoid bony deformity.

Evolving evidence has routinely demonstrated improved accuracy with the use of PSI in shoulder arthroplasty.[8,12,21] In addition, improved component position is expected to increase implant longevity by decreasing the risk of component loosening and failure over time.[12] Given the emergence of next-generation implants with wedge-based version correction, the initial insertion of the central guide pin is critical to reproducing the preoperative plan in the intraoperative setting. Although manufacturer-based custom guides are available, there are several limitations in their adoption for widespread use. These include the cost of the guides, the timing of the CT scan and guide manufacturing before surgery, and the ability to scale production to support widespread routine adoption. The use of in-house, surgeon-generated, 3D printed PSI guides may help to address these issues by providing a quick-turnaround option for guide preparation. In our practice, the guides are typically prepared 1-2 days before the planned procedure, allowing for sterilization and immediate availability during the operation.

Although the results of this study are promising, it has several limitations. Although adequately powered to determine glenoid component accuracy defined as absolute error in version/inclination during aTSA, the study had a small cohort size of 36 total patients, and a larger cohort size would be needed to make more generalizable conclusions. The relatively small cohort size of this RCT made it challenging to perform subgroup comparisons based on the Walch classification of glenoid wear, and a larger study would be better designed to compare PSI with standard instrumentation with 3D templating for each Walch classification subgroup. In addition, despite demonstrating a statistically significant benefit for these patient-specific glenoid drill guides, it is unclear if this benefit is clinically significant. Although a limitation, like the prior RCT by Hendel et al,[8] this study was not designed to define clinical significance and was instead designed to determine if PSI technology could improve glenoid component positioning, particularly in cases with increased glenoid bony deformity. To the best of our knowledge, no study has correlated error in version/inclination correction during aTSA with patient satisfaction to determine clinically significant thresholds for glenoid component placement. Future studies will be needed to better define the clinical benefit of PSI in TSA and better determine the influence of accuracy with glenoid component placement when related to patient satisfaction. Further, the CT scans were obtained at variable time points after surgery from 6 months to 1 year after surgery, and although there was no detectable evidence of implant migration, some subtle cold flow of the implant may have occurred at later time points as demonstrated in an investigation with serial postoperative CT scans and 2-year follow-up by Ricchetti et al.[20] Lastly, the patients in this study had limited abnormalities in glenoid morphology, and future studies are needed to determine the efficacy of 3D printed guides in patients with more severe deformity.

Conclusion

When compared with standard instrumentation, the use of in-house, 3D printed patient-specific glenoid drill guides during aTSA led to more accurate glenoid component version correction and similarly accurate inclination correction. Additional research should examine the efficacy of these guides in more complex settings, such as in severe bony abnormality, reverse TSA, or fractures. Furthermore, the influence of proper component position and use of PSI on clinical outcomes must be further studied.

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